Bo Shang, Ph.D.

Postdoctoral Associate, University of Maryland

Civil and Environmental Engineering / Maryland Transportation Institute

New York, NY

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Robotics and AI researcher focused on reliable multimodal perception and field robotics for transportation and infrastructure. Research connects sensor integration and field data collection with computer vision, embedded control, cross-site model evaluation, and human review of engineering evidence.

Research Interests

Physical AI and field robotics; LiDAR-camera perception; vulnerable-road-user sensing and transportation safety; robotic infrastructure inspection; UAVs, visual servoing, and GPS-denied navigation; reproducible evaluation across sensors, sites, and operating conditions.

Academic and Research Appointments

University of Maryland | Jul 2026 - Present

Postdoctoral Associate, Civil and Environmental Engineering / Maryland Transportation Institute | College Park, MD

- Support hardware and software development, integration, and testing for two sponsored transportation research projects under Dr. Terry Yang.
- Conduct system troubleshooting, support field data collection, and prepare technical documentation and project deliverables.

The City College of New York, CUNY | Jul 2025 - Jun 2026

Postdoctoral Scholar, AI & Mobility Research Lab | New York, NY

- Developed roadside LiDAR and camera workflows for traffic monitoring, multimodal perception, and transportation safety research.

The City College of New York, CUNY | Jan 2025 - Jun 2025

Graduate Research Assistant, Civil Engineering | New York, NY

The City College of New York, CUNY | Dec 2022 - Dec 2024

Postdoctoral Researcher, Robotics Lab | New York, NY

- Developed robotic inspection software, structural-defect analysis, and engineering review workflows connecting sensors, robots, and cloud services.

Vaughn College of Aeronautics and Technology | Jun 2024 - Aug 2024

Instructor, Computer Engineering Summer Academy (AI module) | New York, NY

Vaughn College of Aeronautics and Technology | Jan 2024 - May 2024

Adjunct Faculty | New York, NY

The City College of New York, CUNY | Aug 2023 - Dec 2023

Adjunct Assistant Professor | New York, NY

Missouri University of Science and Technology | Jan 2020 - Nov 2022

Postdoctoral Fellow | Rolla, MO

- Developed UAV guidance and robotic deployment components for bridge inspection and contributed to multimodal infrastructure sensing research.

University of California, Merced | Jan 2016 - Aug 2017

Lecturer | Merced, CA

- Supported teaching in mechatronics, engineering computing, and unmanned aerial systems; developed initial laboratories for the UAS course.

University profile

University of California, Merced | Aug 2015 - Sep 2017
Junior Specialist | Merced, CA

Education

Northeastern University, China | 2013 - 2020
Ph.D., Pattern Recognition and Intelligent Systems

University of California, Merced, USA | 2015 - 2017
Exchange doctoral research

Northeastern University, China | 2011 - 2013
M.Eng., Pattern Recognition and Intelligent Systems

Northeastern University, China | 2007 - 2011
B.Eng., Automation

Research and Engineering Projects

Roadside LiDAR and camera perception, CUNY | Nov 2024 - Jun 2026

- Developed a workflow spanning sensor capture, temporal synchronization, spatial calibration, background subtraction, object segmentation, and road-user detection.
- Built multi-frame vehicle reconstruction, labeling and quality-assurance workflows, and model evaluation across sites with different beam counts, frame rates, and mounting geometries.
- Separated site-specific configuration from reusable processing and investigated cross-configuration generalization for traffic monitoring and vulnerable-road-user safety.

Robotic inspection automation and structural-defect analytics, CCNY | Dec 2022 - Dec 2024

- Developed CNN-based analysis of cracks, spalling, and stains, with AWS inference and a WebODM-based workflow for 3D reconstruction, localization, visualization, and measurement.
- Built an iOS application for controlling a climbing inspection robot and integrated human review into inspection-data workflows.

Bridge Inspection Robot Deployment System (BIRDS), Missouri S&T | 2020 - 2022

- Developed UAV and clamping-robot components for bridge inspection, including vision-based girder detection, embedded perception, and control of the deployment mechanism.
- Integrated NVIDIA Jetson, Raspberry Pi, ROS, embedded Linux, cameras, and robot communication; tested autonomous guidance, clamping, and traversal workflows.

UAV visible, infrared, and hyperspectral sensing | 2020 - 2022
Developed Python and Ubuntu workflows for multimodal infrastructure inspection, sensing, and flight-path planning.

Robot-assisted underwater acoustic imaging for bridge scour evaluation | 2020 - 2021
Contributed Python/C++, ROS, Arduino, embedded control, and computer-vision components to underwater inspection research.

Vision-based UAV servoing and fractional-order control | 2015 - 2019
Developed and evaluated visual guidance and control methods using cameras, embedded Linux, Raspberry Pi, and fractional-order controllers.

SmartCaveDrone and GPS-denied exploration | 2015 - 2017
Contributed RGB-D mapping, visual SLAM, and sense-and-avoid capabilities for UAV exploration and robotic archaeology.

Early robotics projects | 2010 - 2014

- 2013-2014: AR.Drone and ROS quadrotor object tracking.
- 2012-2013: Indoor LiDAR-equipped quadrotor; 2012: camera-equipped quadrotor.
- 2011: Camera-equipped fixed-wing UAV; 2010: infrared line-following smart car.

Independent Research Software and Educational Platforms

ScholarMind AI / Merit Point Academy | 2026 - Present

Independent AI product development

- Developed a bilingual AI education and research platform with a research assistant, course delivery, robotics evidence pages, and knowledge-graph learning tools.
- Directed AI-assisted software development through requirements, implementation, testing, privacy and attribution checks, and deployment.

Reproducible AI evaluation | 2026

Built evaluation workflows using versioned configurations, replay, failure analysis, and regression checks for research software and tool-using AI agents.

Teaching and Mentoring

University of California, Merced | 2016 - 2017

Teaching support for ME 190 Mechatronics, ME 143 Unmanned Aircraft Systems, and ME 021 Engineering Computing (Fortran and MATLAB); initial laboratory development for the UAS course.

The City College of New York, CUNY | 2023

ENGR 204 Electric Circuits, as Adjunct Assistant Professor.

Vaughn College of Aeronautics and Technology | 2024

MCE 355 Robotics, Mechanics and Control; SBC 012 Principles of AI; SBC 014A Principles of Research-AI. Developed and taught a five-lecture, five-lab AI module for the Computer Engineering Summer Academy.

Research mentoring and robotics outreach

Tutored high-school research assistants at CCNY (2023-2024), coached FIRST Robotics Competition participants (2022), and judged VEX Robotics Competition events for middle-school, high-school, and college students (2024).

Journal Articles

[J1] Bo Shang, Yiqiao Li. **Drive-by Urban Freight Sensing: Large-Scale Fine-Grained Truck Classification Using Google Street View Imagery and Deep Learning.** *IEEE Open Journal of Intelligent Transportation Systems, vol. 7, pp. 2327-2347*, 2026. Published September 4, 2026; open access. DOI PDF Record

[J2] Jinglun Feng, Bo Shang, Ejup Hoxha, César Hernández, Yang He, Weihan Wang, Jizhong Xiao. **Robotic Inspection and Data Analytics to Localize and Visualize the Structural Defects of Concrete Infrastructure.** *IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 22324-22336*, 2025. Journal paper selected for presentation at IEEE IROS 2025. DOI Record

[J3] Haibin Zhang, Zhenhua Shi, Liujun Li, Pu Jiao, Bo Shang, Genda Chen. **Code-specified early delamination detection and quantification in a RC bridge deck: passive vs. active infrared thermography.** *Journal of Civil Structural Health Monitoring, vol. 15(1), pp. 227-244*, 2025. Issue year 2025; first published online August 6, 2024. DOI

[J4] Ejup Hoxha, Jinglun Feng, Agnimitra Sengupta, David Kirakosian, Yang He, Bo Shang, Ardian Gjinofci, Jizhong Xiao. **Contrastive learning for robust defect mapping in concrete slabs using impact echo.** *Construction and Building Materials, vol. 461, article 139829*, 2025. DOI SSRN

[J5] Genda Chen, Liujun Li, Haibin Zhang, Zhenhua Shi, Bo Shang. **Aerial Nondestructive Testing and Evaluation (aNDT&E).** *Materials Evaluation, vol. 81(1), pp. 67-73*, 2023. DOI PDF

[J6] B Shang, J Liu, Y Zhang, C Wu, YQ Chen. **Fractional-order flight control of quadrotor UAS on vision-based precision hovering with larger sampling period.** *Nonlinear Dynamics*, 2019.

[J7] B Shang, C Wu, Y Hu, J Yang. **An algorithm of visual reconnaissance path planning for UAVs in complex spaces.** *Journal of Computational Information Systems*, 2014.

[J8] C Gao, C Wu, Y Zhang, B Shang, T Meng. **Research on remote image/data transmission based on TCP/IP protocol.** *Journal of Mechanical & Electrical Engineering, vol. 28(9)*, 2011.

Conference and Workshop Papers

[C1] Bo Shang, Yiqiao Li. **Roadside LiDAR for Cooperative Safety Auditing at Urban Intersections: Toward Auditable V2X Infrastructure Intelligence.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 757-765*, 2026. DriveX Workshop on Foundation Models for Autonomous Driving; archival proceedings. arXiv PDF Record

[C2] Bo Shang, Yiqiao Li, Arian Golrokh Amin, Camille Kamga, Jie Wei. **AI-Enhanced Sensing for Vulnerable Road User Safety at Signalized Intersections: A Survey.** *Procedia Computer Science, vol. 265, pp. 350-357 (MobiSPC 2025)*, 2025. DOI

[C3] Bo Shang, Yiqiao Li, Jie Wei, Camille Kamga. **Investigating the LiDAR Beam Count Needed for Effective Vulnerable Road User Detection.** *Procedia Computer Science, vol. 265, pp. 358-365 (MobiSPC 2025)*, 2025. DOI

[C4] Johnny Li, Bo Shang, Iresh Jayawardana, Genda Chen. **Hardware-in-the-loop and Digital Twin Enabled Autonomous Robotics-assisted Environment Inspection.** *2023 6th International Symposium on Autonomous Systems (ISAS), pp. 1-5*, 2023. DOI PDF

[C5] Liujun Li, Genda Chen, Bo Shang. **Mixed Reality Enabled Digital Twin for Robot-Assisted Bridge Element Inspection and Maintenance.** *8th World Conference on Structural Control and Monitoring (8WCSCM)*, 2022.

[C6] Z Shi, B Shang, H Zhang, L Li, G Chen. **Evaluation of User-Friendliness of Several UASs for Bridge Inspection.** *8th World Conference on Structural Control and Monitoring (8WCSCM)*, 2022.

[C7] H Zhang, P Jiao, L Li, Z Shi, B Shang, G Chen. **Delamination Detection of Concrete Bridge Slab Through UAV-Based Thermal Scanning.** *8th World Conference on Structural Control and Monitoring (8WCSCM)*, 2022.

[C8] Pu Jiao, Bo Shang, Liujun Li, Genda Chen. **The ceiling effect and flight insight of unmanned aerial vehicles during proximity inspection of bridges via computational fluid dynamics modeling and simulations.** *Structural Health Monitoring 2021: Proceedings of the 13th International Workshop on Structural Health Monitoring, pp. 866-874*, 2021. Proceedings year 2021; workshop held March 15-17, 2022. DOI Record

[C9] B Shang, C Wu, YQ Chen. **Neighborhood Optimization Method for Shaping Bode Plot With Larger Phase Margin.** *ASME IDETC/CIE*, 2019.

[C10] B Shang, C Wu, Y Zhang, YQ Chen. **Fractional order flight control of quadrotor UAS: a Simscape benchmark environment and a case study.** *IEEE CSAA Guidance, Navigation and Control Conference (CGNCC)*, 2018.

[C11] B Shang, Y Zhang, C Wu, YQ Chen. **Fractional order flight control of quadrotor UAS: an OS4 benchmark environment and a case study.** *15th International Conference on Control, Automation, Robotics and Vision (ICARCV)*, 2018.

[C12] G Zhang, B Shang, YQ Chen, H Moyes. **SmartCaveDrone: 3D cave mapping using UAVs as robotic co-archaeologists.** *International Conference on Unmanned Aircraft Systems (ICUAS)*, 2017.

[C13] B Shang, C Wu, Y Zhang, YQ Chen. **Analysis of maximum possible sampling period for a real-time vision-based control system.** *ASME IDETC/CIE*, 2017.

[C14] B Shang, J Liu, T Zhao, YQ Chen. **Fractional order robust visual servoing control of a quadrotor UAV with larger sampling period.** *International Conference on Unmanned Aircraft Systems (ICUAS)*, 2016.

[C15] B Shang, C Wu, T Meng, C Gao, Y Zhang. **A Data/Image Transmission Device Based on TCP/IP Protocol.** *8th International Conference on Wireless Communications, Networking and Mobile Computing (WiCOM)*, 2012.

[C16] T Meng, C Wu, B Shang, C Gao, Y Zhang. **Design of point to multi-point wireless communication system based on ZigBee.** *7th International Conference on Wireless Communications, Networking and Mobile Computing (WiCOM)*, 2011.

Annual Meeting Contributions

[M1] F Arnob, Bo Shang, Yiqiao Li, Abolfazl Karimpour. **Understanding Infrastructure-ADAS Interactions in Urban Crash Risk: A Multimodal AI Approach for Intersection Safety.** *TRB 105th Annual Meeting (Transportation Research Board)*, 2026. Record

[M2] Bo Shang, Yiqiao Li, Blanche Horbach, Kevin Villon. **Scalable Urban Truck Characterization using Deep Learning and Street View Imagery: A Drive-by Sensing Approach.** *TRB 105th Annual Meeting (Transportation Research Board)*, 2026. Record

[M3] Yiqiao Li, Jie Wei, Bo Shang, Camille Kamga. **Few-Shot Learning for Truck Classification from LiDAR Sensor using Vision-language Models.** *TRB 105th Annual Meeting (Transportation Research Board)*, 2026. Record

Preprints

[P1] Yiqiao Li, Bo Shang, Jie Wei. **Bridging the Modality Gap in Roadside LiDAR: A Training-Free Vision-Language Model Framework for Vehicle Classification.** *arXiv preprint arXiv:2602.09425*, 2026. arXiv DOI

[P2] Bo Shang, Yiqiao Li, Blanche Horbach, Kevin Villon. **Exploring Drive-By Sensing and Deep Learning for Urban Truck Characterization Using Google Street View Imagery.** *SSRN preprint 5312573*, 2025. DOI SSRN

Technical Reports

[R1] Yiqiao Li, Bo Shang, Weicong Feng, Jie Wei, Camille Kamga. **Investigation of Emerging Sensing and AI/ML Technologies to Enhance the Safety of Vulnerable Roadway Users at Signalized Intersection.** *U.S. Department of Transportation, University Transportation Centers Program, final technical report*, 2026. Published May 1, 2026; National Transportation Library record DOT 91974. PDF Record

[R2] Pengfei Ma, Tarutal Ghosh Mondal, Ying Zhuo, Zhenhua Shi, Bo Shang, Liujun Li, Genda Chen. **CAAP Final Report.** *Pipeline and Hazardous Materials Safety Administration (PHMSA), CAAP Final Report*, 2022. September 20, 2022; contract 693JK31950005CAAP; 109 pp. Record

[R3] Y Hu, J Yang, Y Cao, F Yang, B Shang. **Northeastern University Autonomous Aerial Robotics Team.** *International Aerial Robotics Competition (IARC) Symposium*, 2014.

[R4] Y Hu, J Yang, B Shang, S Liu. **Northeastern University Autonomous Aerial Robotics Team.** *International Aerial Robotics Competition (IARC) Symposium*, 2013.

Theses

[T1] Bo Shang. **Fractional Order Proportional Derivative Visual Servoing Control for UAVs.** *Northeastern University, China (Ph.D. dissertation)*, 2020. In Chinese; 132 pp. DOI

[T2] B Shang. **SLAM Research Based on ROS and Indoor Quadrotor.** *Northeastern University (M.S. Thesis)*, 2013.

Posters and Presentations

Posters

[PO1] B Shang, L Li, P Jiao, R Huerta, J Ressel, A Rawlings, B Scharfenberg, et al.. **Drone vision-based clamping strategy for bridge inspection.** *INSPIRE-UTC Poster*, 2021. Record

[PO2] B Shang, A Reven, P Jiao, B Li, G Chen. **Vision-Based Non-GPS UAV Guidance for Bridge Inspection.** *INSPIRE-UTC 2020 Annual Meeting (Poster)*, 2020.

[PO3] A Reven, P Jiao, B Shang, G Chen. **Clamping Design for Bridge Inspection Robot Deployment Systems (BIRDS) Prototype II.** *INSPIRE-UTC 2020 Annual Meeting (Poster)*, 2020.

Presentation Slides

[S1] A Reven, P Jiao, B Shang, G Chen. **Bridge Inspection Robot Deployment Systems (BIRDS) Prototype II.** *INSPIRE-UTC 2020 Annual Meeting (Slides)*, 2020.

Patents and Patent Applications

[PT1] Genda Chen, Alec Reven, Bo Shang, Zhenhua Shi, Liujun Li, Boyang Li. **Unmanned vehicle having flight configuration and surface traverse configuration.** *U.S. Patent 12,296,994 B2*, 2025. Granted May 13, 2025. Patent

[PT2] Chengdong Wu, Bo Shang, Yunzhou Zhang, Chengxi Gao, Tingting Meng. **Data/image transmission device based on TCP/IP (Transmission Control Protocol/Internet Protocol).** *Chinese Patent CN 102427464 B*, 2014. Granted April 30, 2014. Patent

[PT3] Yunzhou Zhang, Bo Shang, Chengdong Wu, Pengju Si. **Internet-based interactive digital media terminal device.** *Chinese Patent Application CN 102306237 A*, 2012. Patent application published January 4, 2012. Patent

Professional Service

Journal reviewer

Nonlinear Dynamics; Journal of Intelligent & Robotic Systems; ISA Transactions; IEEE Transactions on Control Systems Technology; Intelligent Buildings International; International Journal of Advanced Robotic Systems; Control Engineering Practice; IET Control Theory and Applications; Mechatronics.

Conference reviewer

International Conference on Unmanned Aircraft Systems (ICUAS); IEEE International Conference on Robotics and Automation (ICRA); IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI).

Session Chair | 2016

International Conference on Unmanned Aircraft Systems (ICUAS).

Review record

Web of Science profile

Honors, Support, and Credentials

Teaching Certificate, Association of College and University Educators (ACUE) | 2021

FAA Remote Pilot Certificate for Small Unmanned Aircraft Systems | 2016

China Scholarship Council (CSC) doctoral exchange support | 2015 - 2017

Two-year exchange program at the University of California, Merced; US$38,400 in financial support.

International Aerial Robotics Competition | 2014

Best System Control Award and Best Mission Planning Award, as team leader.

International Mathematical Contest in Modeling | 2010

Meritorious Prize, as programmer.

National Smart Car Competition, Freescale, China | 2010

First Prize, Northeastern Region.

Technical Expertise

Robotics and sensing

ROS / ROS 2, NVIDIA Jetson, Raspberry Pi, Arduino, embedded Linux; LiDAR, RGB/RGB-D, thermal, IMU, and acoustic sensing; sensor bring-up, synchronization, calibration, and field troubleshooting.

Software and machine learning

Python, C++, PyTorch, TensorFlow, OpenCV, Linux, Docker, Git/GitHub Actions, AWS; computer vision, 3D reconstruction, visual SLAM, model packaging, and data pipelines.

Evaluation and research delivery

Dataset and configuration versioning, labeling and data quality assurance, cross-site evaluation, replay and failure analysis, human review, GIS and traffic analytics, and technical documentation.